

Baby Cries Project

Stats 141XP: Lecture 1 Group 1
Justin Noche, Naina Sharma, Josh Xu,
Minjae Kim, Cynthia Du, Daniel Jenkins
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Introduction

- It has been claimed that babies cry uniquely for any and every reason, including diaper changes and hunger.
- 1250 observations were collected on an application that records audio files of baby cries. Parents were then asked to label the type of cry based on their best judgement.
- This project aims to discover whether there are any discernible categories of baby cries based on the recorded audio data.
- After an exploration of the demographic population of the babies, our team used unsupervised learning to discover how many cry types exist in the data given the acoustic features. Then, supervised learning was performed given the parent labels to find cross-validated accuracy.

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Summary Statistics of Demographics



ID

Unique ID of each baby. 499 recorded with NO-EMAIL
Some babies are recorded more than once.



Reason

Five Reasons with 500 Observations each:
Diaper-Change, Fussy, Hungry, Pain, and Tired



Age

Age given in days.
Some recorded as MULT-REDCAP-AGE, NO-AGE, -1 and 0.



Gender

950 female and 944 male observations.
Some recorded as MULT-REDCAP-GENDER, or NO-GENDER

Summary Statistics of Demographics

ID:

- NO-EMAIL entries had NO-AGE and NO-GENDER. 499 Recorded with NO-EMAIL

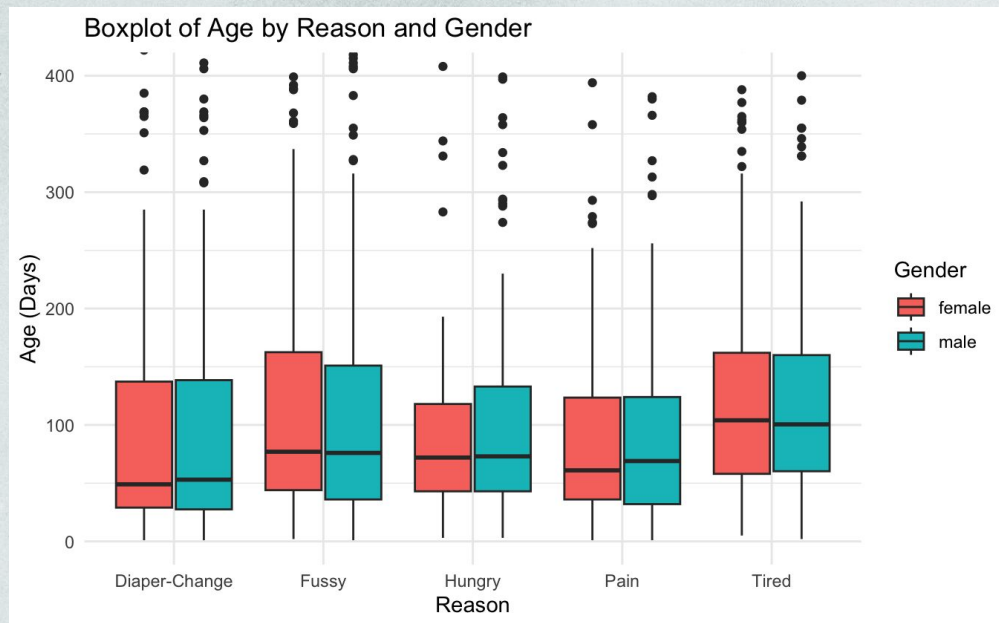
Age:

-1	0	NO-AGE	MULT-REDCAP
3	732	7	2

0 - 12 Months	1 - 3 Years	3+ Years
1630	11	15

Gender:

Female	Male	NO-GENDER	MULT-REDCAP
950	944	604	2



Using cleaned age observations, including outliers. Plot limit at 400 days for visual purposes.

Statistical Analysis

Boxplot of Age by Reason and Gender

- The boxplot shows that the Tired Category has the highest median age for both genders.
- The long tails shows that there is potential right skewing.

ANOVA

- Does Age differ between Reason levels?
→ **Yes** ($p\text{-value} < 0.05$)

Chi-Squared Test

- Is there an association between Gender and Reason
→ **No**

Filtered Acoustic Dataset

- 1250 Individual recorded baby cries given
- Raw audio data was transformed into 25 acoustic features with 8 categories such as pitch, formant, and loudness

F0semitoneFrom27.5Hz_sma3nz_pctlrange0.2	alphaRatioV_sma3nz_stddevNorm	StddevUnvoicedSegmentLength	loudness_sma3_stddevNorm
5.167824	-2.190032	0.2522026	1.217534
2.302433	4.403415	0.775806	1.036633
16.07217	-0.7259132	0.8472849	0.812052
6.519093	1.314629	0.3047228	0.8059498
4.542534	-3.228497	0.1083205	0.6468157
1.203712	7.8158	0.1975035	0.5271363
13.47438	1.23272	1.138977	1.219144
5.818726	-1.624396	0.2238722	0.8423291
6.009083	1.168851	0.1233584	0.5847681
3.712673	2.269837	0.08407965	0.5834538

Unsupervised Clustering

Data Preprocessing

- Kept all observations with acoustic variables ($n = 1250$), discarded Gender and Age
- combined/deleted highly correlated variables
 - Variables related to 20th and 80th percentiles were transformed into range variables

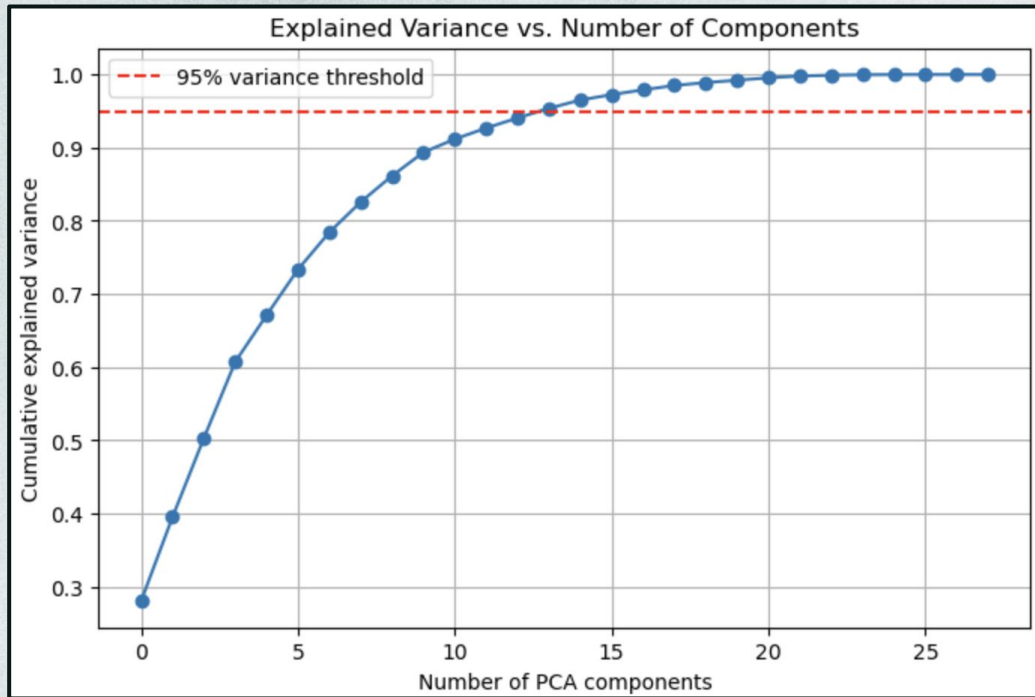
Dimensionality Reduction

- Principal Component Analysis
- Linear Discriminant Analysis
- UMAP
- t-SNE

Clustering Algorithms

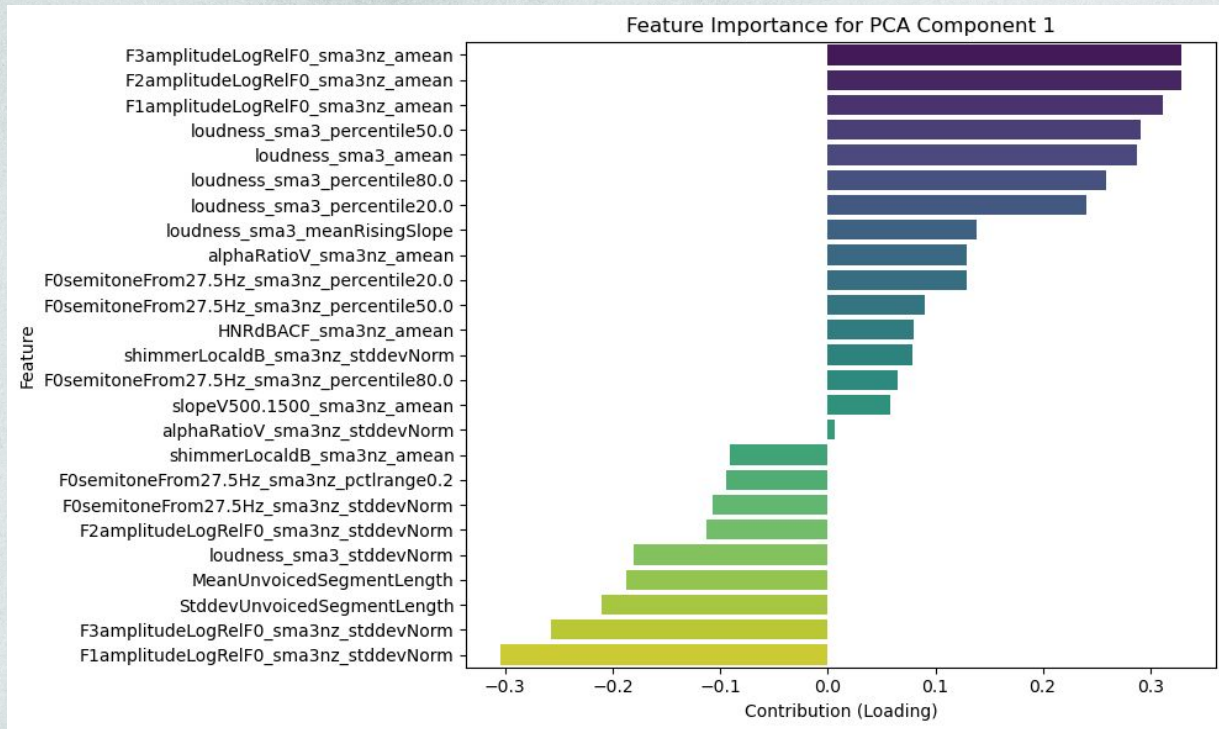
- K-Means Clustering
- Gaussian Mixture Models
- Agglomerative Clustering
- DBScan

PCA - Dimensions of Cry Acoustics



To capture 95% of the variance in the cry acoustic data, we need **13 principal components**

PCA - Analysis of Principal Components

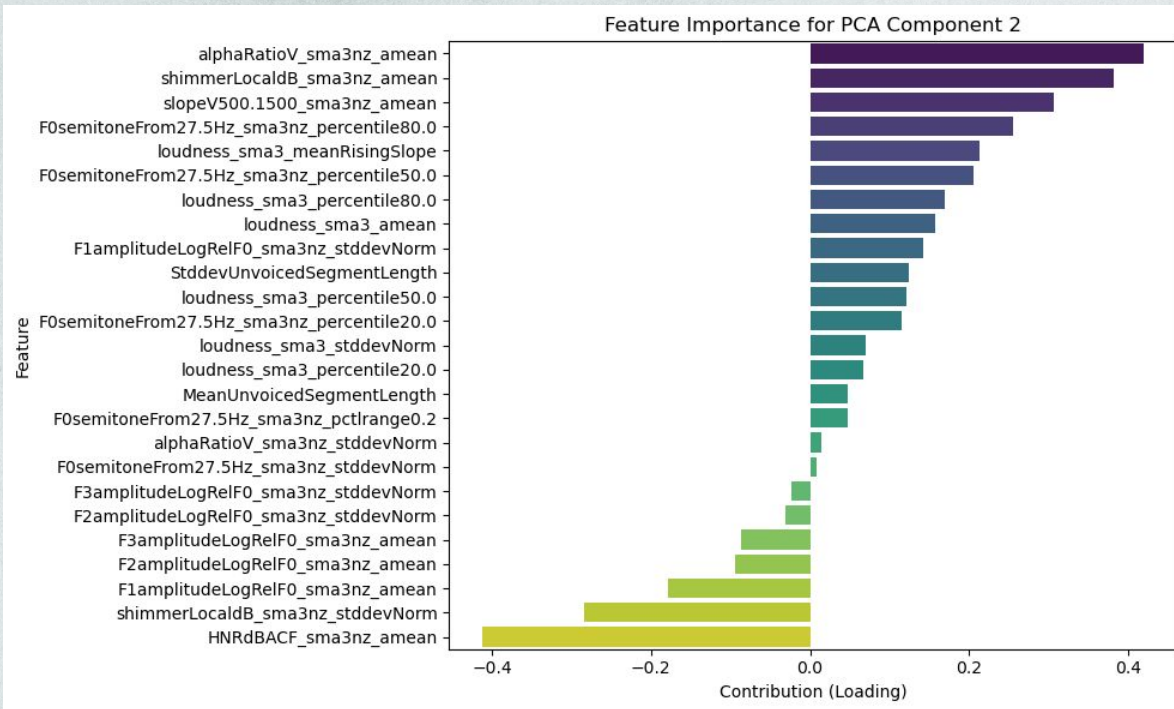


First principal component is primarily driven by

- **Formant amplitudes** (F1, F2, F3 relative to pitch), which reflect resonance characteristics of the cry
- **Loudness characteristics**, such as average and median intensity
- **Temporal variability**, particularly in unvoiced (silent) segments

These results suggest that the **vocal tract shape and loudness dynamics** are the strongest acoustic indicators of different cry types.

PCA - Analysis of Principal Components



Principal Component 2 (PC2) is primarily driven by

- **Voice quality measures:**

Harmonics-to-noise ratio and shimmer (both mean and variability)

- **Spectral features:** Alpha ratio (timbre) and spectral slope

- **Pitch and loudness dynamics**

These features likely reflect **how clear or noisy a baby's cry sounds**, potentially distinguishing cries like hunger (clear, tonal) from pain or tiredness (breathy, strained). This supports the idea that spectral and voice-quality cues play a key role in differentiating emotional states in infant cries.

PCA – Analysis of Principal Components

3rd Principal Component

- Low, median, and high pitch percentiles,
- Pitch variability and range,
- Harmonics-to-noise ratio and shimmer (voice clarity),
- Duration and variability of unvoiced (silent) segments.

This component likely reflects the **pitch contour and melodic structure of a cry**. Cries that rise and fall more dramatically in pitch may score high here, which could differentiate urgent or distressed cries (e.g., pain) from more stable cries (e.g., hunger or tiredness)

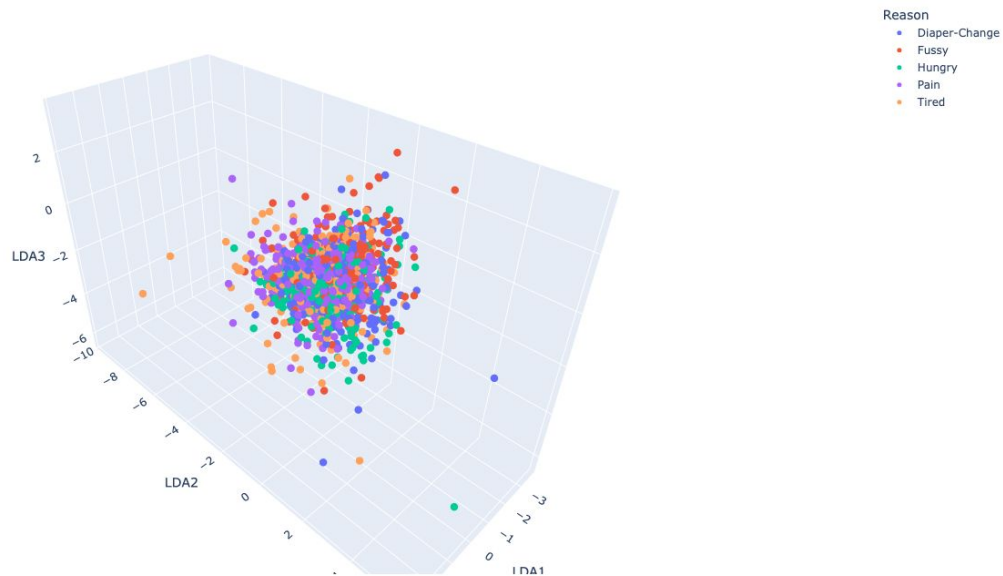
4th Principal Component

- Formant variability, particularly F2 and F3 amplitudes relative to pitch (reflecting vocal tract resonance changes),
- Loudness dynamics, including rising slope, overall amplitude, and loudness percentiles (80th, 50th),
- Amplitude irregularities, such as shimmer variability,
- Spectral shape, via spectral slope.

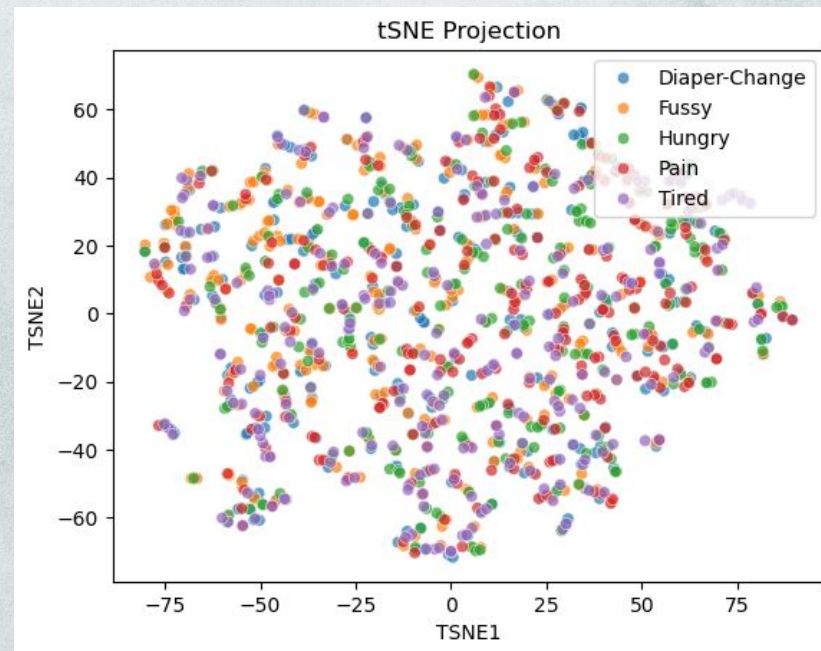
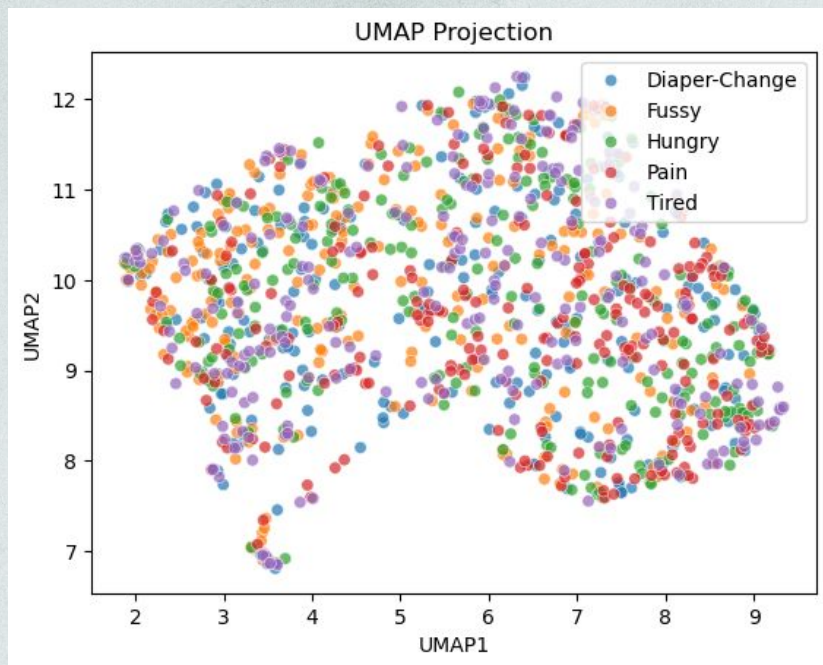
This component likely captures the **energy modulation and resonance variability in a baby's cry**. Cries with strong fluctuations in loudness and resonance — such as escalating or high-arousal cries, may load high on this component. It could help distinguish cries with more expressive or strained delivery from more stable or flat cries

LDA 3D Projection

LDA 3D Projection (Interactive)

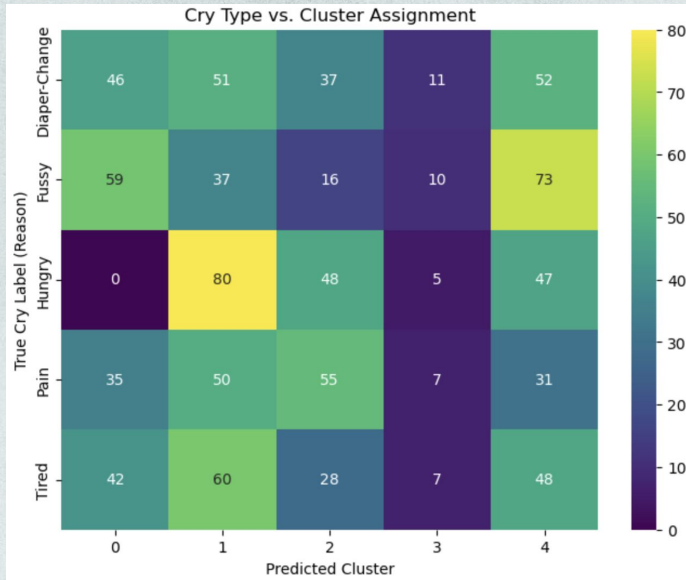


UMAP & t-SNE Projections



KMeans with 5 Clusters

Without PCA



0: Fussy, 1: Hungry, 2: Pain, 3: Diaper, 4: Fussy

With PCA



0: Fussy, 1: Diaper, 2: Hungry, 3: Fussy, 4: Pain

Gaussian Mixture Model



Uniqueness Score = count in most common cluster / total count for that cry type

Hungry: 12, 152, 6, 5, 5 $\rightarrow 152 / 180 = \mathbf{84.4\%}$

Fussy: 9, 66, 10, 105, 5 $\rightarrow 105 / 195 = \mathbf{53.8\%}$

Pain: 17, 77, 18, 64, 2 $\rightarrow 77 / 178 = \mathbf{43.3\%}$

Tired: 7, 78, 11, 84, 5 $\rightarrow 84 / 185 = \mathbf{45.4\%}$

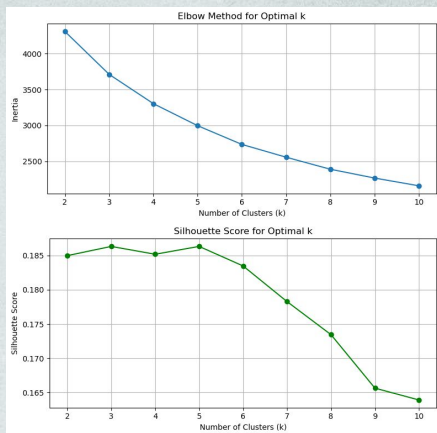
Diaper-Change: 15, 74, 17, 82, 9 $\rightarrow 82 / 197 = \mathbf{41.6\%}$

Ranking (Most to Least Separable):

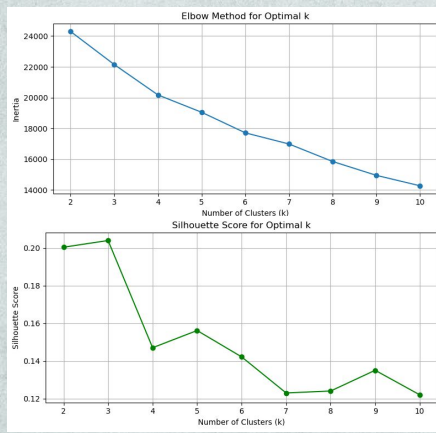
1. **Hungry** – 84.4% in Cluster 1
2. **Fussy** – 53.8% in Cluster 3
3. **Tired** – 45.4% in Cluster 3
4. **Pain** – 43.3% in Cluster 1
5. **Diaper-Change** – 41.6% in Cluster 3

Finding the Optimal Number of Clusters

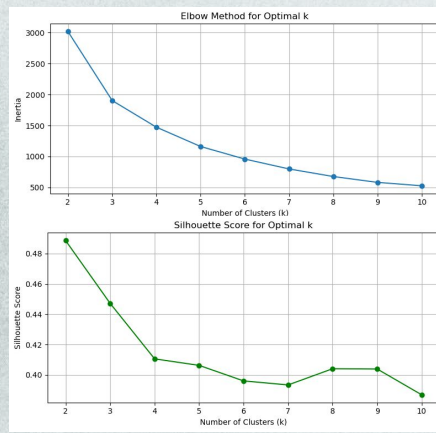
K-Means on LDA...



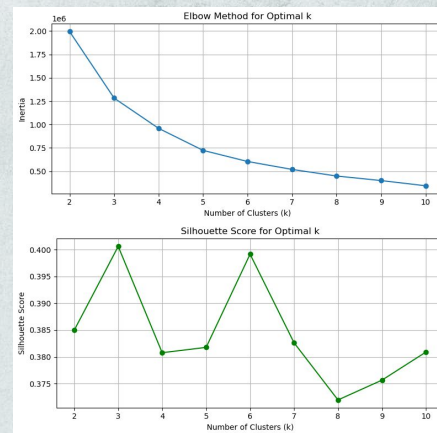
K-Means on PCA...



K-Means on UMAP...

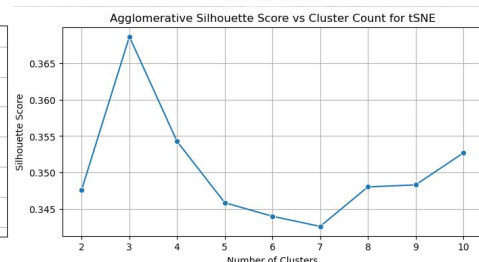
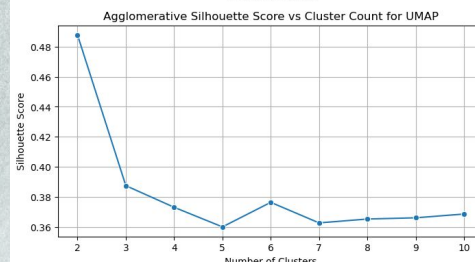
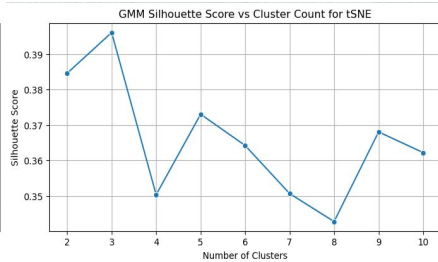
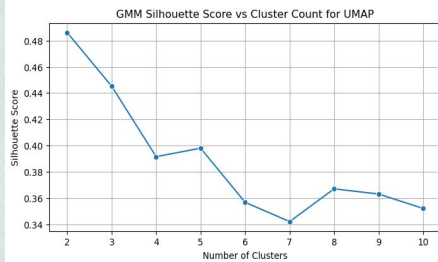
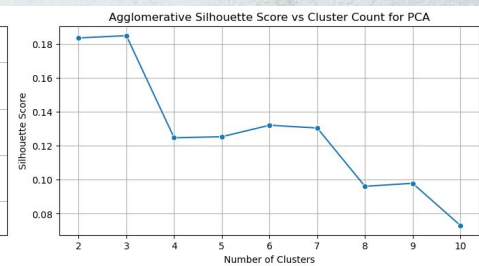
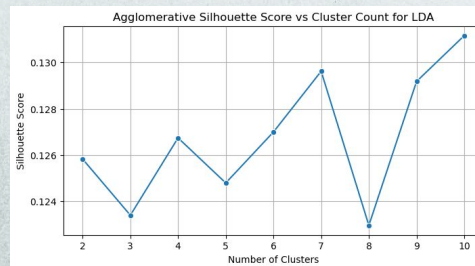
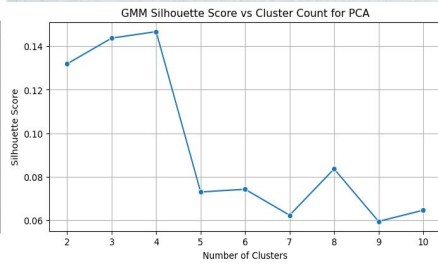
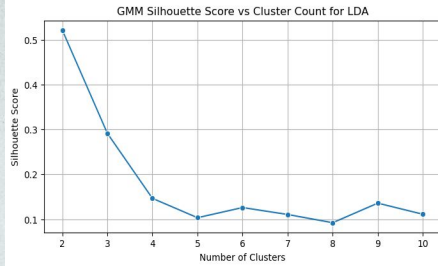


K-Means on t-SNE...

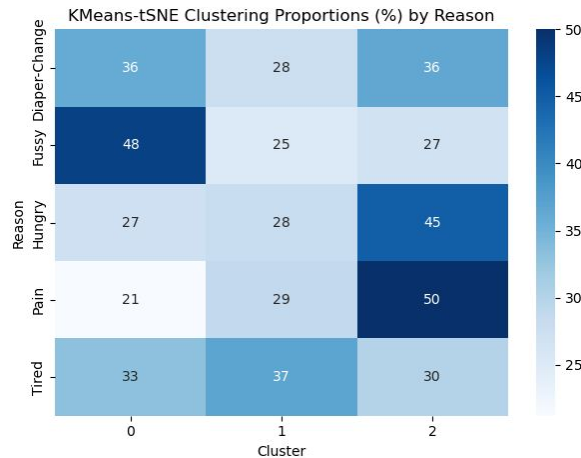
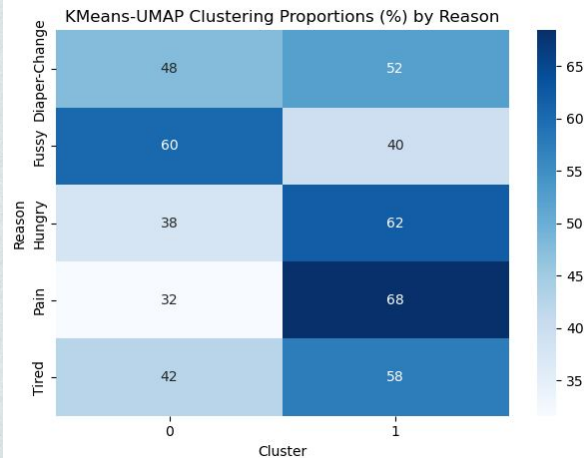
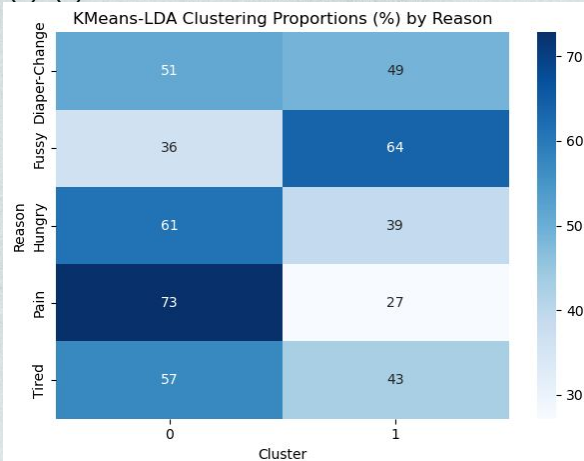


Conclusion: K-Means consistently gave either $k = 2$ or $k = 3$ for the optimal number of clusters as by silhouette scores

Finding the Optimal Number of Clusters – GMM & Agglomerative

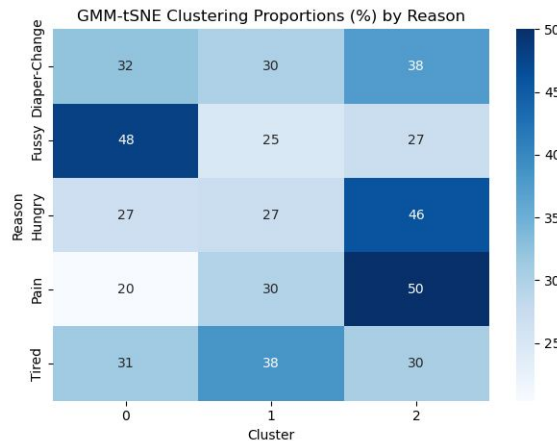
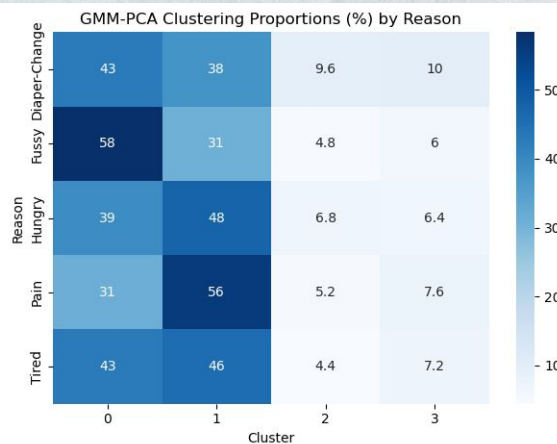
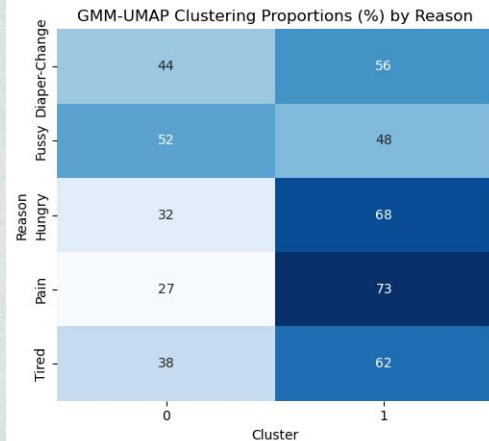
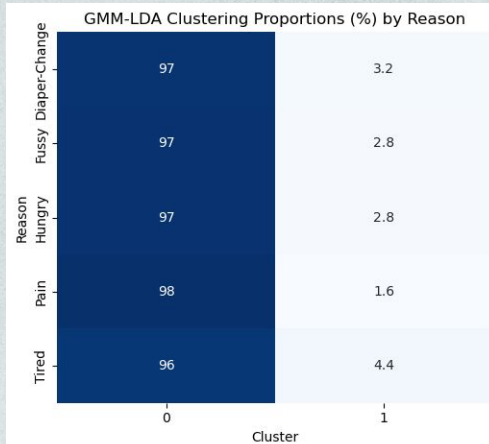


Conclusion: Both GMM and Agglomerative Clustering gave $k = 2$ or $k = 3$ as the optimal number of clusters with the exception of some datasets (ex. Agglomerative Clustering on LDA gave $k = 10$)



KMeans Results

- K-Means Clustering gave either 2 or 3 clusters, successfully mapping fussy cries to one cluster and tired, pain, hungry cries to the other cluster
- K-Means Clustering on tSNE dimensionality reduction shows that pain and hungry cries were similar to each other, and fussy cries were the most distinct

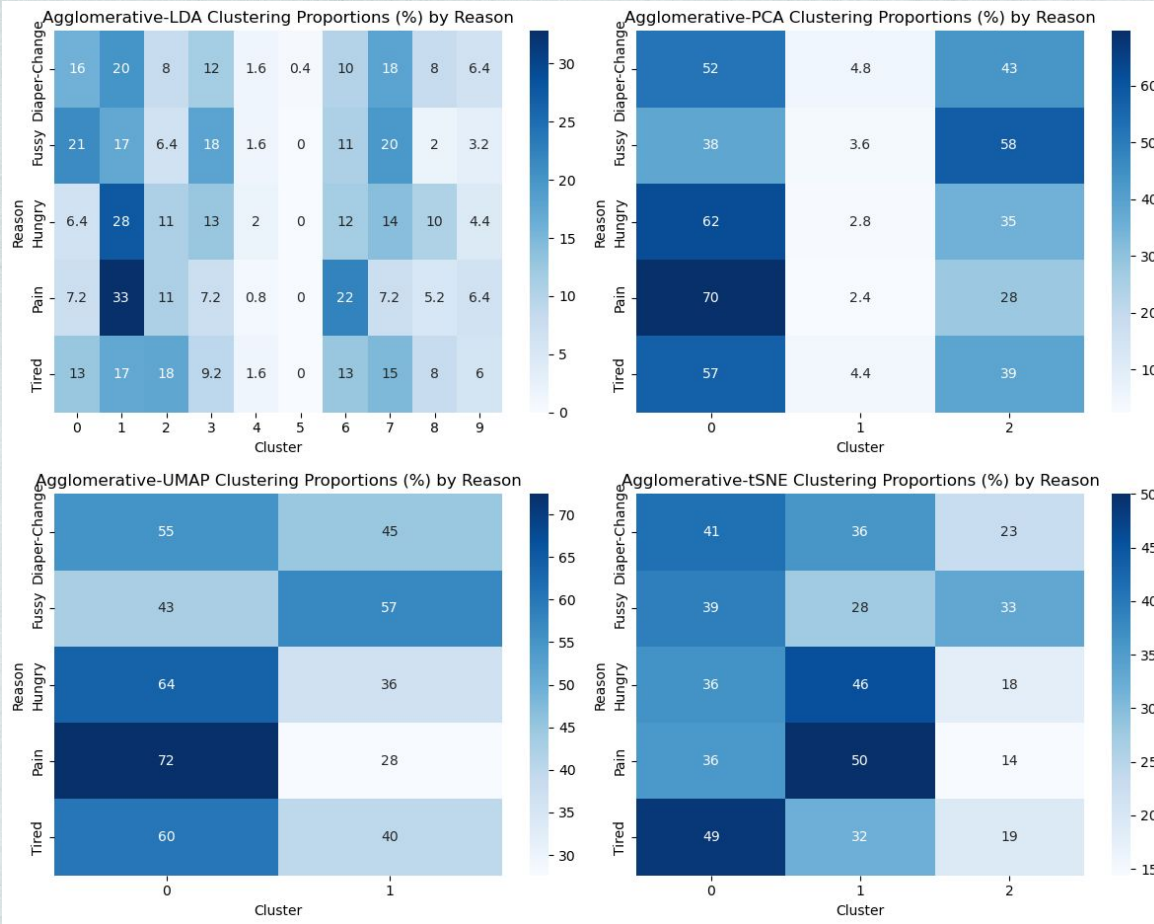


GMM Results

- GMM showed a marked increase in performance for UMAP and tSNE dimensionality reduction
- As with K-Means Clustering, GMM also classified tired, pain and hungry cries mostly into the same cluster, while classifying fussy into the other cluster
- GMM was unsuccessful in classifying LDA reduced data

Agglomerative Clustering Results

- Agglomerative Clustering also shows an increased performance when clustering the data points into either 2 or 3 clusters
- Agglomerative Clustering on tSNE was able to somewhat distinguish hungry and pain cries from tired cries
- Heatmaps on LDA and PCA show that increasing the number of clusters does not signify an increase in clustering performance. Rather, these clusters were selected as 'optimal' because they most likely contained a subset of outliers that were mapped close together



Unsupervised Learning Conclusion

How many cry types/clusters exist in the data based on acoustic features?

- There appear to be 2-3 distinct cry patterns, even though the parents labeled 5 reasons

How many dimensions are the cry acoustics?

- PCA concluded there were 13 principal components for the cry acoustics

When the data is clustered into categories, how well do these categories map to parent labels (confusion matrix)?

- See earlier slides!

Which type of cry is the most separable and unique? Rank categories from most to least unique/separable.

- Fussy cries were the most separable, with the other cluster mostly mapping to pain, hungry or tired cries

Supervised Learning – Parent Labels

- **K-Nearest Neighbors: simple**, non-parametric algorithm that classifies a data point based on the majority class of its K closest neighbors in the feature space
 - Accuracy = 23.8%
- **Random Forest**: method that builds multiple decision trees and combines their outputs to improve prediction accuracy and reduce overfitting
 - Accuracy = 24.8%
- **Multinomial regression**: generalization of logistic regression used for multi-class classification problems, estimating probabilities for each class
 - Accuracy = 25.6%
- **Support vector machine**: classification model that finds the optimal hyperplane to separate classes by maximizing the margin between them; effective in high-dimensional spaces
 - Accuracy = 33.2%
- ★ **Gradient Boost (XGboost)**: ensemble technique that builds models sequentially, with each new model correcting errors from the previous one; highly effective for structured data
 - Accuracy = 62.56%

k-Nearest Neighbors

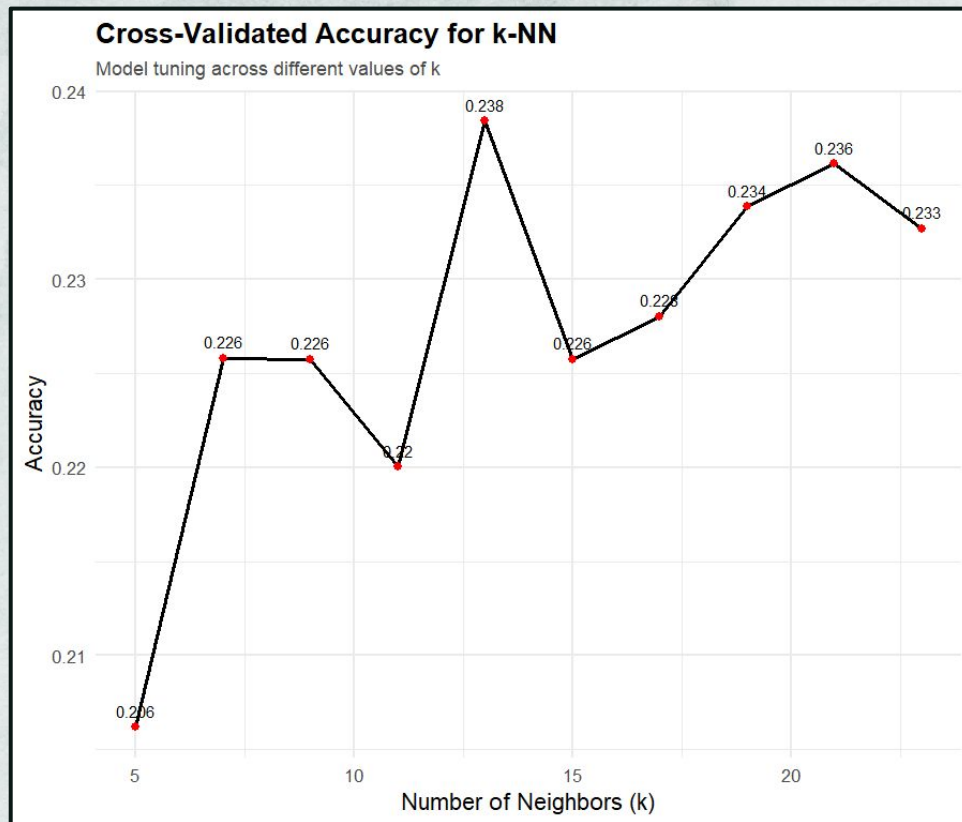
“Weak Separation of
Vocalization Categories”

Best Accuracy: 23.8% at $k = 13$

Expected by chance: ~20%

Kappas: < 0.06

Acoustic features alone may
not reliably differentiate labeled
categories using kNN



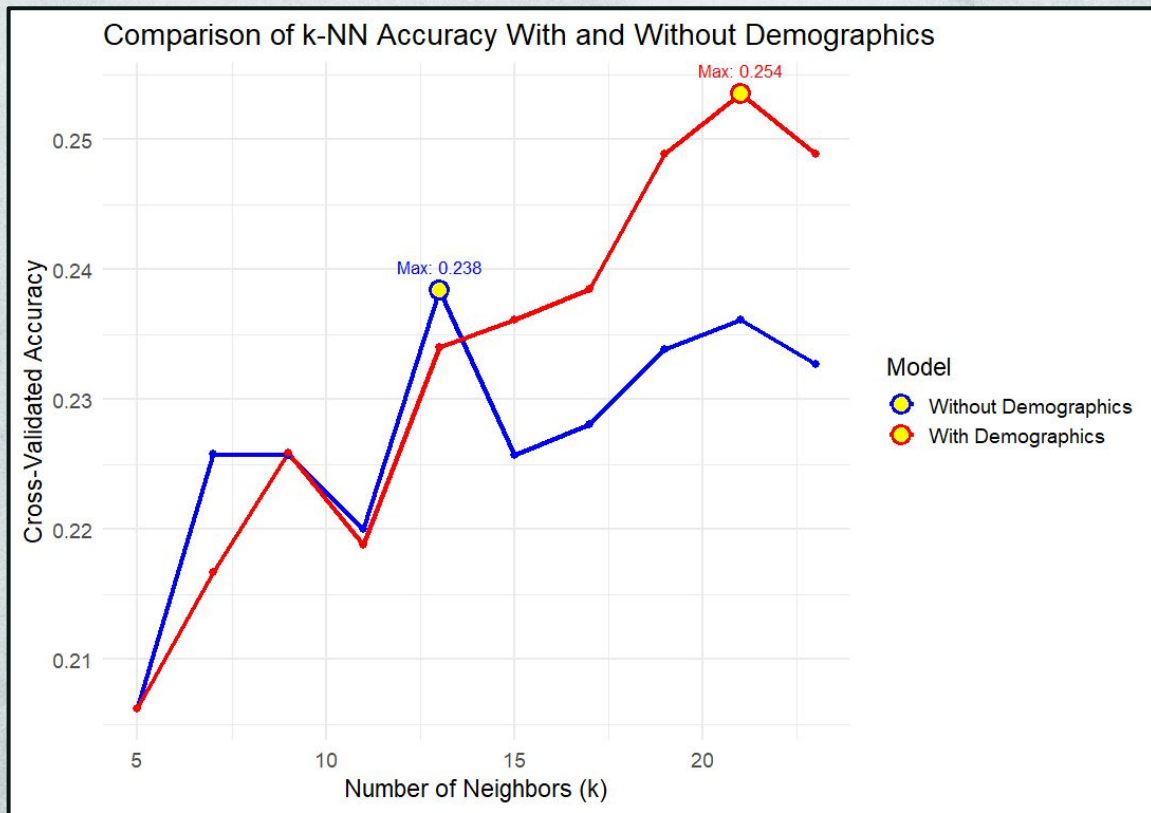
Adding Demographic Info to k-NN

k-NN tested *with* vs. *without* **age**
and **gender**

Accuracy slightly **improves** when
adding demographics

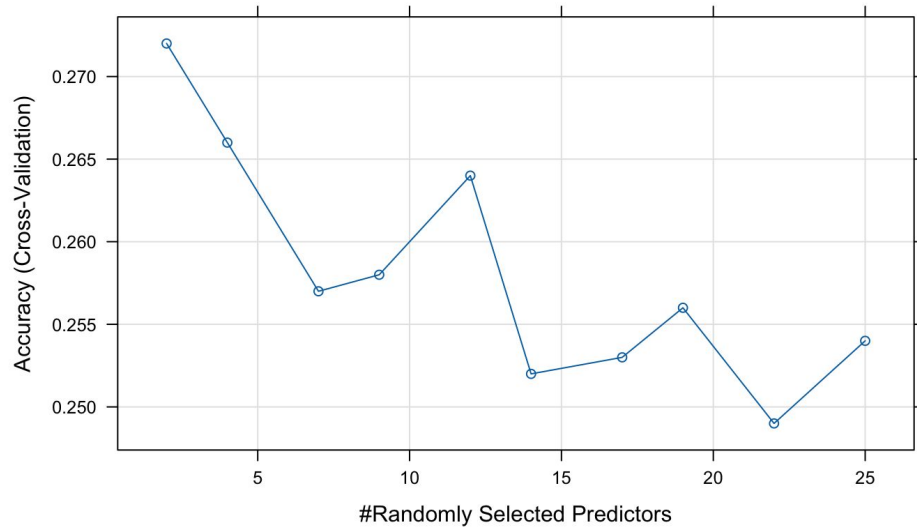
Model	Best Accuracy	Best k
With age & gender	25.4%	21
Only acoustics	23.8%	13

Smoother class boundaries



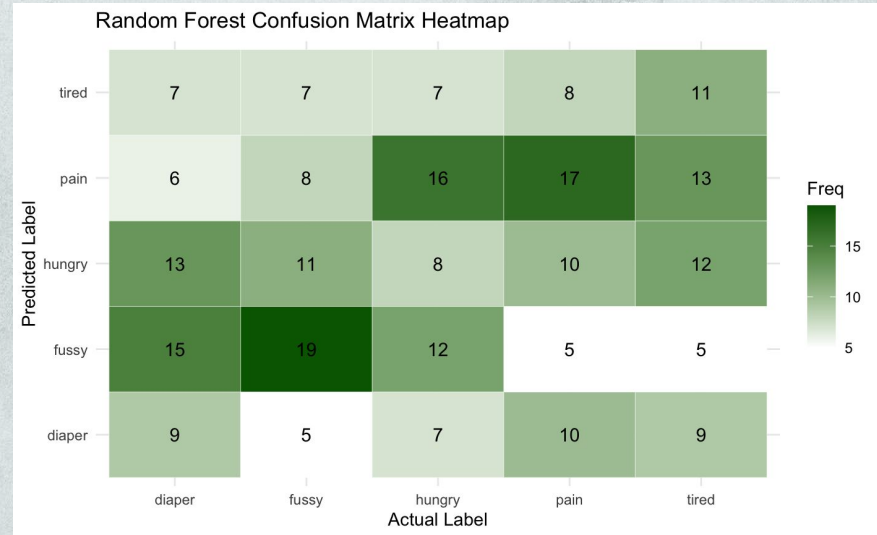
Random Forest

predictors vs accuracy



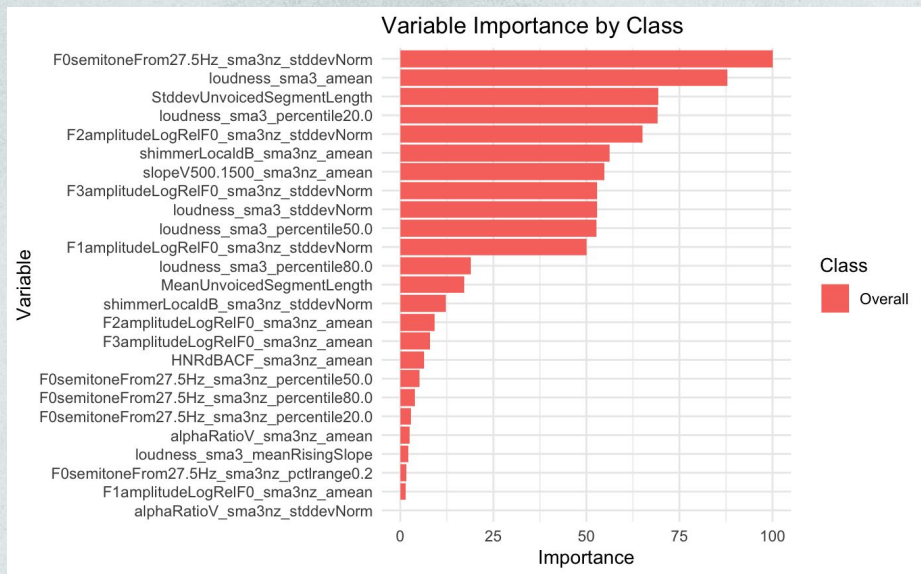
This plot shows the number of features randomly selected in the RF model vs the cross validated accuracy. We ran the model with $mtry = 2$, and $mtry = 12$, and $mtry = 25$ to see which yielded the best results. $mtry = 12$ and $mtry = 25$ yielded the same accuracy.

This heatmap visualizes how well the Multinomial model correctly classifies each category. Darker diagonal tiles indicate better performance, while off-diagonal values show misclassifications. The model was slightly better at predicting fussy and pain labels, but generally did not have great predictive ability.



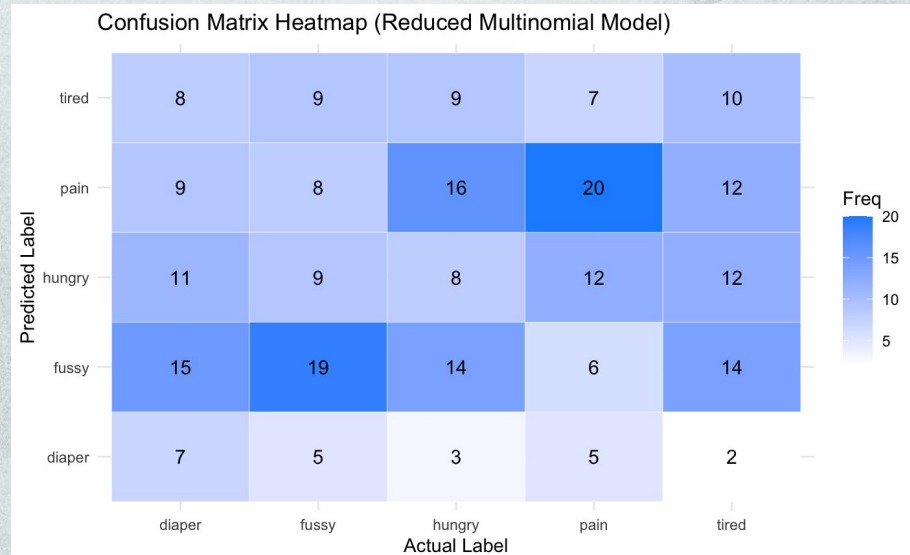
Total accuracy = 25.6%

Multinomial Regression



This plot shows how influential each predictor is in distinguishing between outcome classes in the multinomial model. Variables with higher importance contribute more to the model's decision-making across all categories. Based on the results, we used 11 features in our reduced model.

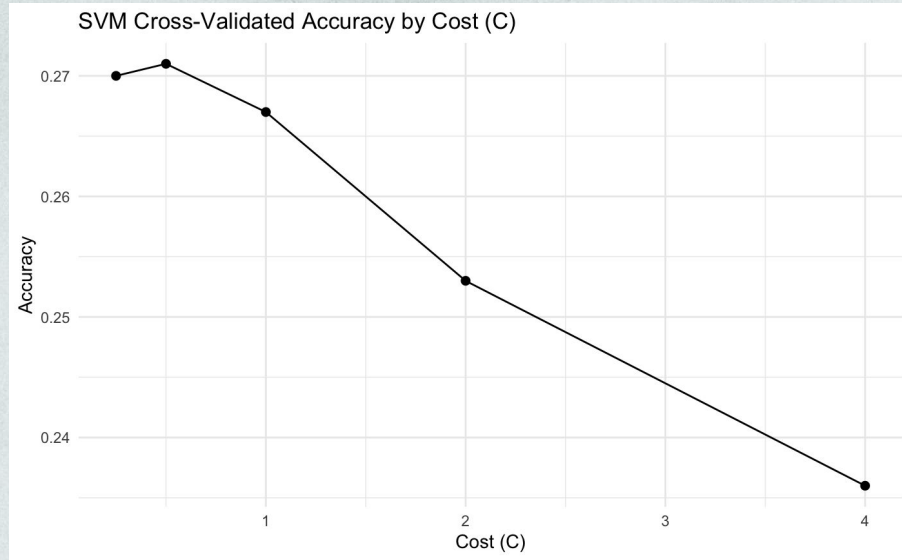
This heatmap visualizes how well the multinomial model correctly classifies each category. Darker diagonal tiles indicate better performance, while off-diagonal values show misclassifications.



Total accuracy of reduced model = 25.6%
accuracy did not change between reduced and full multinomial model

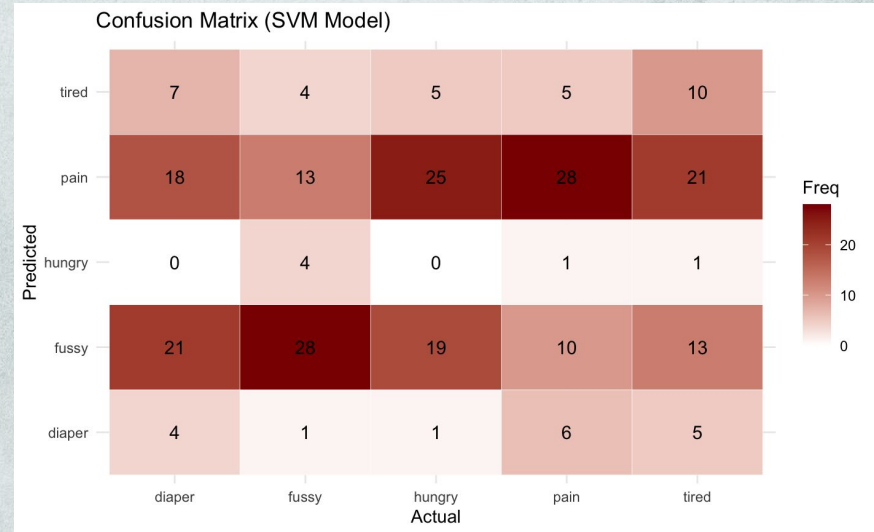
Support Vector Machine

Cost vs Cross Validated Accuracy



This plot shows the model's accuracy at different values of the cost parameter (C), helping identify the optimal balance between bias and variance

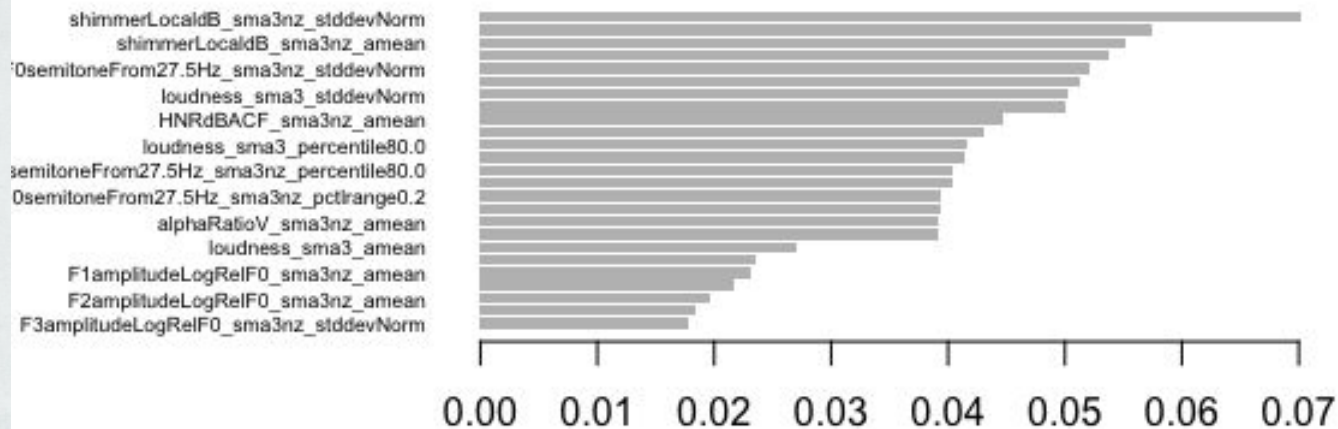
This heatmap visualizes how well the SVM model correctly classifies each category. While the model classified pain and fussy relatively well, it generally did not have good predictive ability.



Total accuracy = 33.2%

XGBoost

XGBoost Variable Importance Plot



Model accuracy increased to 98% with tuned parameters when adding demographic information to the dataset.

Supervised Learning - Cluster Labels

Using cluster labels of N categories, what is the average cross-validated accuracy?

Process:

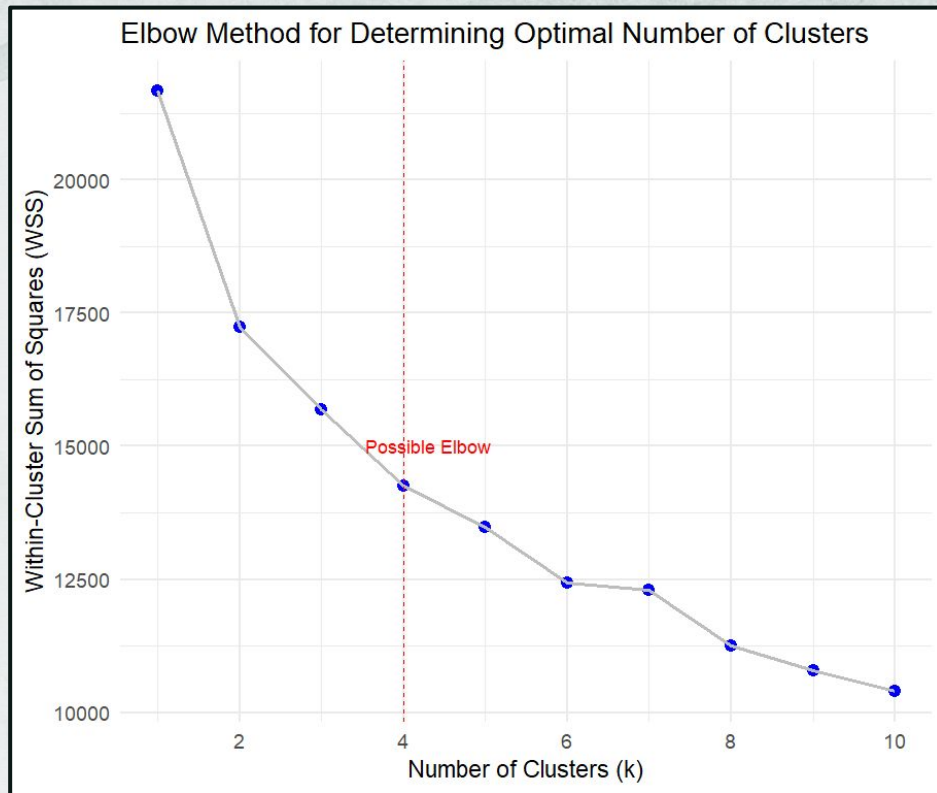
1. Clean dataset
2. Run k-means clustering (unsupervised) to obtain cluster labels
 - a. Use elbow and silhouette methods to determine optimal number of clusters
3. Train supervised models using 5 fold cross validation:
 - a. Random forest
 - b. Support Vector Machine
 - c. K-Nearest Neighbors
 - d. XGBoost
4. Evaluate the performance of each model
5. Repeat with demographic data to see if it improves the classification accuracy

Elbow Method for Optimal Groups

Need for “best”
number of k clusters

WSS (within-cluster
sum of squares) =
Measure of closeness
of **point to center** of
cluster.

Elbow: where
improvements level off



Optimal
“ $k = 4$ ”

Use of $k = 3$

Odd number of groups
avoids ties.

3 groups more distinct
than 5

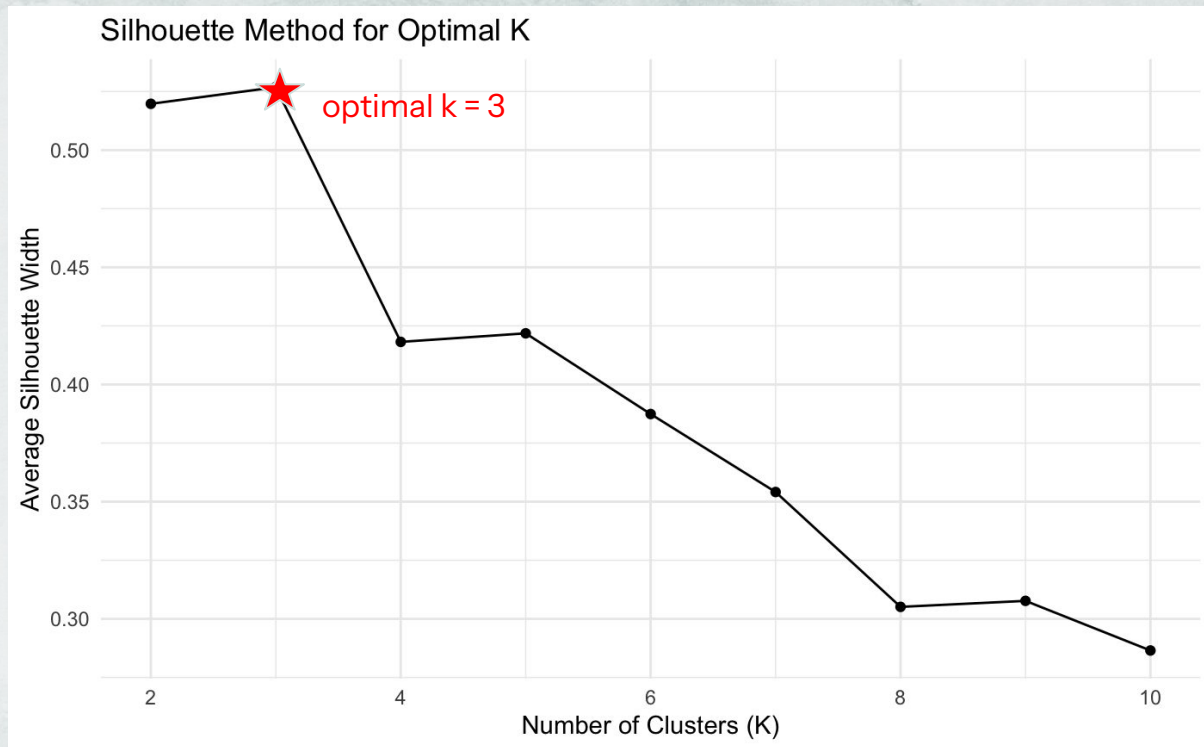
Silhouette Method for Optimal Groups

Silhouette method: measuring how well each observation fits within its cluster

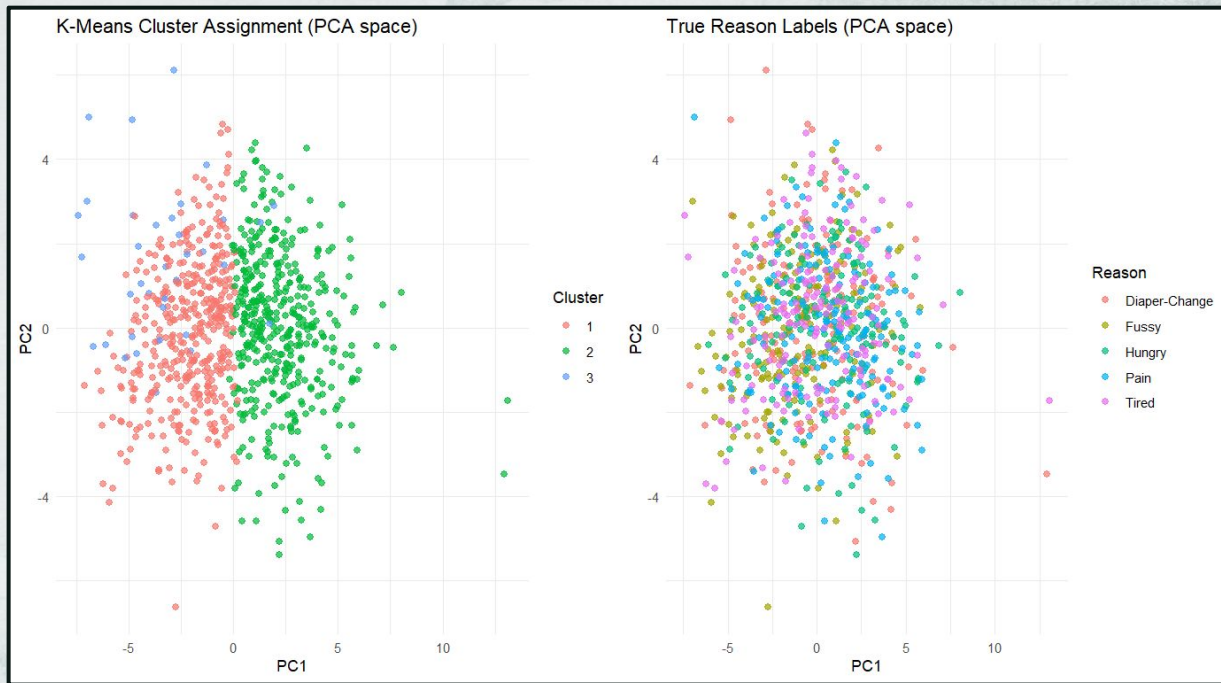
For each K , computes silhouette width $s(i) = a$

Scores range from -1 to 1:

- Close to 1: well-clustered
- Around 0: ambiguous
- Below 0: likely in the wrong cluster



K-Means Clusters to True Values



Random Cluster Labels:

Clear

Distinct

True Reason Labels:

Mixed

Overlapping

Acoustic features **don't fully**
capture reason distinction

Labeled reasons noisy

Model results using $k = 3$ clusters

Method	Parameters	Cross Validated Accuracy
Random forest	mtry = 13	98.56190%
K-Nearest Neighbors	k = 9	98.39742%
Support vector machine	sigma = 0.04265, c = 1	93.28284%
XGBoost	nrounds = 100, max_depth = 3, eta = 0.3, gamma = 0, colsample_bytree = 0.8, min_child_weight = 1, subsample = 0.5	98.31966%

Adding demographic information

Process:

1. Use combined dataset, remove non feature columns
2. Remove observations with no age or gender
3. Recode gender = 1 if female and 0 if male as kmeans only takes numeric predictors
4. Convert `age` to numeric
5. Apply kmeans clustering on the cleaned data to get cluster labels using $k = 3$
6. Use resulting cluster labels as pseudo-outcomes to train and evaluate models with 5-fold CV:
 - a. Random forest
 - b. K-Nearest Neighbors
 - c. Support Vector Machine
 - d. XGBoost
7. Record accuracy and best parameters for each model

Model Results with Demographic Info

Method	Parameters	Cross Validated Accuracy
Random forest	mtry = 14	99.77011%
K-Nearest Neighbors	k = 7	99.65517%
Support vector machine	sigma = 0.03488, c = 1	99.42396%
XGBoost	nrounds = 50, max_depth = 1, eta = 0.3, gamma = 0, colsample_bytree = 0.6, min_child_weight = 1, subsample = 1	99.77011%

Supervised Learning Conclusion

Using 5 parent labels, what is the average cross-validated accuracy?

- Gradient boost model with all the features yields the highest accuracy of 63%
- Random forest, SVM, and multinomial regression all performed slightly better than random chance → weak models

Using cluster labels of N categories, what is the average cross-validated accuracy?

- Cluster labels performed significantly better than using parent labels
- The highest average cross validated accuracy is 98.56% using a Random Forest model with 3 clusters

Does adding demographic information to the classifier improve accuracy?

- Using parent labels, adding demographic information did not move the needle with accuracy
- Using cluster labels with 3 categories, adding demographic information did improve accuracy slightly



Thank You!

